

DESIGN DOCUMENT



FORCE
FALL · RATE · ALERT

Fall & Optical Rate Cardiac Emitter

PLATFORM

ESP32-WROOM-32

DATE

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Heart Rate & HRV

SpO₂

Body Temperature

Fall Detection

Alerts & Safety

OBJECTIVES

Minimum Viable Product



HR accuracy ± 5 bpm vs. clinical reference (resting & active)



SpO₂ accuracy $\pm 2\%$ vs. finger pulse oximeter (85–100% range)



Temperature accuracy ± 0.5 °C



Fall detection sensitivity $\geq 95\%$ on validated fall dataset



Power management, resulting in 48h battery life



Alert latency < 10 s

Add-Ons



Form factor: compact and ergonomic



All data encrypted in transit



Hot-swappable battery



Water & Dust resistance: IP44

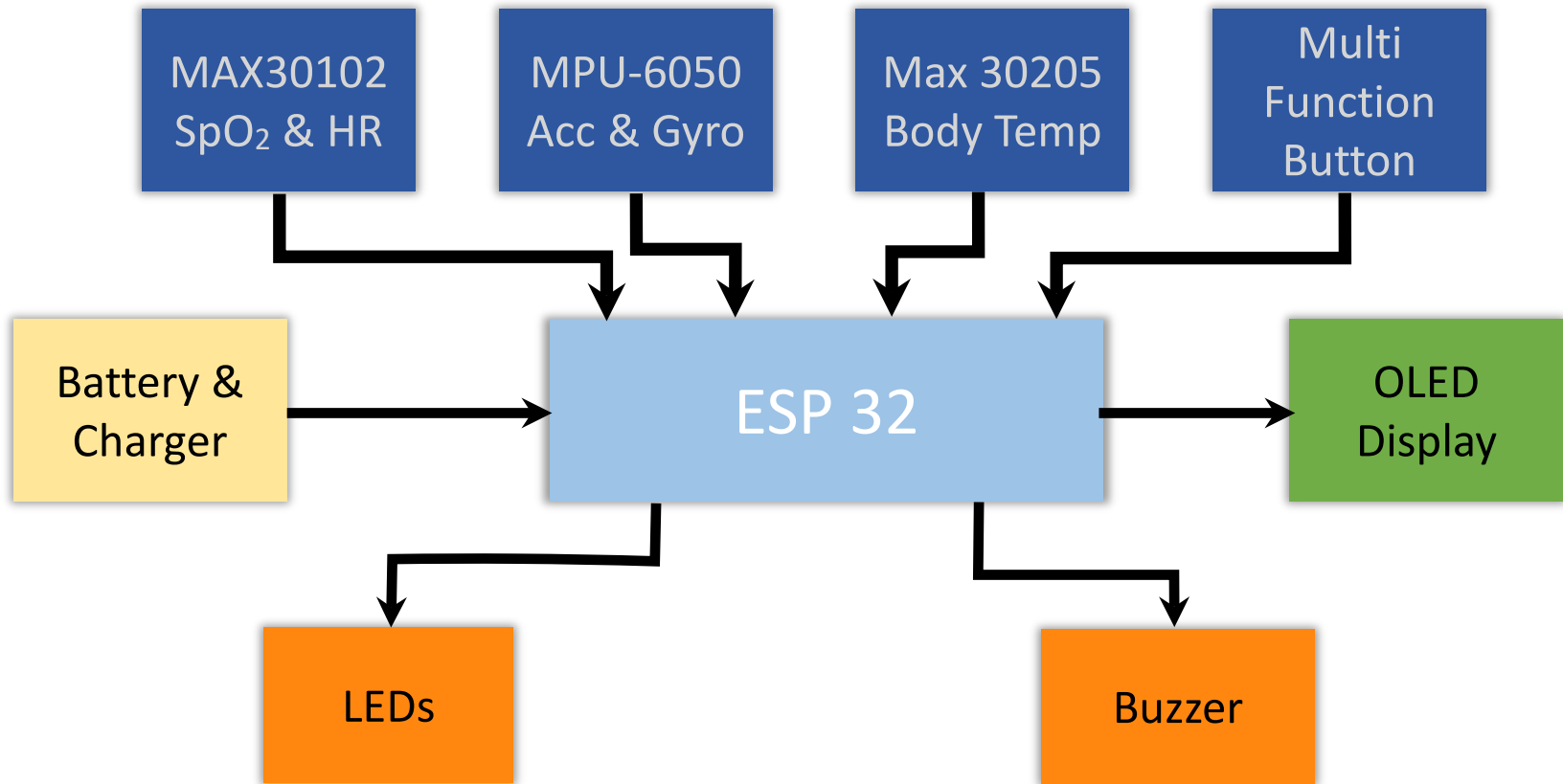


One week battery life on provided lithium cell



Panic Button

HARDWARE ARCHITECTURE DIAGRAM



SOFTWARE COMPONENTS

FORCE is a bicep-worn, band-like device for elderly users that continuously monitors vital signs and detects anomalies and dangerous falls — generating real-time local alerts and, optionally, if provided with a WiFi connection, forwards the alerts to caretakers.



Vitals Monitoring

Processes data coming from the available sensors and checks against the baseline and medical guidelines.

SpO₂

HR

Tmp.



Fall Detection

Six-axis sensor fusion + Heart rate variability. Two-stage validation algorithm reduces false positives.

Acc.

Gyro

HR



Safety & Alerts

Onboard buzzer + RGB LED for local alerts; Display for details.

OPTIONAL: Caretaker alert pipeline and WiFi comms with backend for statistics.

Disp.

Buzz.

Butt.



Power Management

Deep-sleep scheduling, sensor duty cycles and dynamic polling rates optimise battery life for long periods of continuous use.

Batt.

Chrg.